MENG-4227

Failure of Mechanical Components

01

Introduction to Failure Analysis

Comprehensive Study Notes

Course Instructor

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Chapter 1: Introduction

Mode of Fracture → Loading Type

Fracture Mechanism → Fracture Microscopic Process

Failure analysis deals with the determination of the causes of the failure of engineering parts or components. Failure can be defined as the inability of a component to function properly, and this definition does not imply fracture. Failure analysis can be defined as the examination of a failed component and of the failure situation in order to determine the causes of the failure. The purpose of a failure analysis is to define the mechanism and causes of the failure.

The causes of failures can be broken down into the following categories:

1. **Misuse**. The component is placed under conditions for which it was not designed.
2. **Assembly errors and improper maintenance**. Assembly errors involve such factors as leaving off a bolt or using incorrect lubricant. Maintenance of equipment ranges from painting surfaces to cleaning and lubrication, and its neglect may lead to failure.
3. **Design errors**. This is a very common cause of failure. In this category the following items are considered to be specified by the design process:
   a) **Size and shape of the part**. This is usually determined by stress analysis or geometric constraints.
   b) **Material**. This refers to the chemical composition and the treatment (for example, heat treatment) necessary to achieve the required properties.
   c) **Properties**. This is related to stress analysis, but other properties such as corrosion resistance must also be considered.

It is interesting to examine some information about the causes of failures:

1. **Improper material selection** (38%)
2. **Improper maintenance** (44%: Aircraft Components)
3. **Faulty design considerations**. Causes of failures due to faulty design considerations or misapplication of material include the following:
   a) Ductile failure (excess deformation, elastic, or plastic; tearing or shear fracture)
   b) Brittle fracture (from flaw or stress raiser of critical size)
   c) Fatigue failure (load cycling, strain cycling, thermal cycling, corrosion fatigue, rolling contact fatigue, fretting fatigue)
   d) High-temperature failure (creep, oxidation, local melting, warping)
   e) Static delayed fractures (hydrogen embrittlement, caustic embrittlement, environmentally stimulated slow growth of flaws)
   f) Excessively severe stress raisers inherent in the design
   g) Inadequate stress analysis, or impossibility of a rational stress calculation in a complex part

   h) Mistake in designing on the basis of static tensile properties, instead of the significant material properties that measure the resistance of the material to each possible failure mode.
4. **Faulty processing**. Causes of failures due to faulty processing include the following:
   a) Flaws due to faulty composition (inclusions, embrittling impurities, wrong material)
   b) Defects originating in ingot making and casting (segregation, unsoundness, porosity, pipes, nonmetallic inclusions)
   c) Defects due to working (laps, seams, shatter cracks, hot-short splits, delamination, excess local plastic deformation)
   d) Irregularities and mistakes due to machining, grinding, or stamping (gouges, burns, tearing, fins, cracks, embrittlement)
   e) Defects due to welding (porosity, undercuts, cracks, residual stress, lack of penetration, under bead cracking, heat-affected zone)
   f) Abnormalities due to heat treating (overheating, burning, quench cracking, grain growth, excessive retained austenite, decarburization, precipitation)
   g) Flaws due to case hardening (intergranular carbides, soft core, wrong heat cycles)
   h) Careless assembly (such as mismatch of mating parts, entrained dirt or abrasive, residual stress, gouges, or injury to parts)
   i) Parting-line failures in forging due to poor transverse properties.
5. **Deterioration in service**. Causes of failures due to deterioration during service conditions include the following:
   a) Overload or unforeseen loading conditions
   b) Wear (erosion, galling, seizing, gouging, cavitation)
   c) Corrosion (including chemical attack, stress corrosion, corrosion fatigue, dezincification, graphitization of cast iron, contamination by atmosphere)
   d) Inadequate or misdirected maintenance or improper repair (such as welding, grinding, punching holes, cold straightening)
   e) Disintegration due to chemical attack or attack by liquid metals or plating at elevated temperatures.
   f) Radiation damage (sometimes must decontaminate for examination, which may destroy vital evidence of cause of failure); varies with time, temperature, environment, and dosage.
   g) Accidental conditions (such as abnormal operating temperatures, severe vibration, sonic vibrations, impact or unforeseen collisions, ablation, thermal shock).

The steps involved in conducting a failure analysis and their sequence depend upon the failure. One sequence to do this includes the following eight steps:

1. **Description of the failure situation**. Here the history of the failure should be documented. Any information pertaining to the failure, such as the design of the component (including the material and properties), and how the component was being used, is important to obtain. Especially useful are photographs of the part and of associated components.
2. **Visual examination**. Here the general appearance of the part should be documented. Care should be exercised in handling the part so as not to damage any of the fracture surfaces or other important features.
3. **Mechanical design analysis (stress analysis)**. When the part clearly involved mechanical design as a major design component, a stress analysis should be carried out. This will help to establish whether the part was of sufficient size and of proper shape, and what mechanical properties were required. In some cases, this analysis may establish the cause of failure. For example, if the load on a part can be determined and estimates of the mechanical properties made, then it may be possible to establish that the part is too small for this load.
4. **Chemical design analysis**. This step refers to an examination of the suitability of the material from the standpoint of corrosion resistance.
5. **Fractography**. Examination of the fracture surface with the unaided eye, with optical microscopes, and with electron microscopes should be carried out in order to establish the mechanism of fracture.
6. **Metallographic examination**. This requires sectioning and metallographic preparation. It may require agreement between all parties involved before sectioning. This step will help to establish such facts as whether the part had the correct heat treatment.
7. **Properties**. The properties pertinent to the design should be determined. This is not always possible because the test to determine a property may destroy the part. In terms of mechanical properties, hardness is especially important. Hardness will frequently correlate with many other mechanical properties (such as yield strength). It is a simple test to perform, and it usually will not damage the part.
8. **Failure simulation**. A very useful approach is to take an identical (supposedly) part and subject it to the exact condition under which it is designed to operate. This may be too expensive to carry out and is not done frequently.

An alternative procedure for metallurgical failure analysis:

1. Collection of background data and selection of samples
2. Preliminary examination of the failed part (visual examination and record keeping)
3. Nondestructive testing
4. Mechanical testing (including hardness and toughness testing)
5. Selection, identification, preservation, and/or cleaning of all specimens
6. Macroscopic examination and analysis (fracture surfaces, secondary cracks, and other surface phenomena)
   EX/ Shaft Fracture due to Torsion: 1. Smearing (shiny rotated surface at the edge) 2. Color Bands
   EX/ Cup and Cone Fracture: 1. Shear Lip (outer diameter) 2. Fibrous (fibers are formed by the stretching and thinning)
7. Microscopic examination and analysis
   EX/ Polymers Fracture (Mode: Brittle – Thermosetting Polymers) → Fibrillar Bridges (Crazing) → Micro Voids → Crack
   EX/ Tensile Failure of Composites: like-pull out radial marks on the fractured fibre surfaces
   EX/ Ceramics: Internal Void → Crack propagation from the ceramic coping towards the veneering porcelain
8. Selection and preparation of metallographic sections
   - The surface of a metallographic specimen is prepared by various methods of grinding, polishing, and etching.
   - Metallographic specimens are mounted using a hot compression thermosetting resin. Mounting a specimen provides a safe, standardized, and ergonomic way by which to hold a sample during the grinding and polishing operations.
   - After mounting, the specimen is wet ground to reveal the surface of the metal.
9. Examination and analysis of metallographic sections
   EX/ Incipient melting in micro fractography refers to a condition where a small amount of the material being tested has been heated to the point of melting but then rapidly cooled back to its solid state due to the surrounding material acting as a heat sink.
10. Determination of failure mechanism
11. Chemical analyses (bulk, local, surface corrosion products, deposits or coatings, and microprobe analysis)
    EX/ Hydrogen Embrittlement (Steel): Hydrogen embrittlement is a permanent loss of strength that can occur in some steels when hydrogen atoms are present, and stress is applied. Embrittlement occurs as hydrogen atoms migrate to the region of highest stress and cause microcracks. When a crack forms, the hydrogen then migrates to the tip of the crack and causes continued crack growth until the effective cross-section is so reduced that the remaining cross-section is overloaded.
    Micro-Feature: coarse granular morphology and intergranular cracks (dark lines)
12. Analysis of fracture mechanics

13. Testing under simulated service conditions (special tests)
14. Analysis of all the evidence, formulation of conclusions, and writing the report (including recommendations)

Determination of the mechanical properties frequently plays a prominent role in failure analysis, and these properties can often be estimated from hardness measurements. This procedure is essentially nondestructive.

Fractography

Fractography is the interpretation of features observed on fracture surfaces in order to establish the cause of failure. This may be due to stress overload, fatigue, creep, stress corrosion or some combination of mechanisms, depending on the conditions.

Fractography → Macro/Micro-Scopic → 1. Load Type 2. Failure Mode

Failure Mechanisms

1. Fatigue Failures

Metal fatigue is caused by repeated cycling of the load. It is a progressive localized damage due to fluctuating stresses and strains on the material. Metal fatigue cracks initiate and propagate in regions where the strain is most severe.

The process of fatigue consists of three stages:

1. Origin, crack initiation
2. Progressive crack growth across the part
3. Final sudden fracture of the remaining cross section

2. Wear Failures

Wear may be defined as damage to a solid surface caused by the removal or displacement of material by the mechanical action of a contacting solid, liquid, or gas. It may cause significant surface damage and the damage is usually thought of as gradual deterioration. While the terminology of wear is unresolved, the following categories are commonly used. Common types include:

- Adhesive wear
- Abrasive wear
- Erosive wear

- Cavitation Wear
- Fatigue Wear

Adhesive wear has been commonly identified by the terms galling or seizing. Abrasive wear, or abrasion, is caused by the displacement of material from a solid surface due to hard particles or protuberances sliding along the surface. Erosion, or erosive wear, is the loss of material from a solid surface due to relative motion in contact with a fluid that contains solid particles. More than one mechanism can be responsible for the wear observed on a particular part.

3. Corrosion Failures

Corrosion is chemically induced damage to a material that results in deterioration of the material and its properties. This may result in failure of the component. Several factors should be considered during a failure analysis to determine the affect corrosion played in a failure.

Type of corrosion

1. Corrosion rate
2. The extent of the corrosion
3. Interaction between corrosion and other failure mechanisms

Corrosion is a normal, natural process. Corrosion can seldom be totally prevented, but it can be minimized or controlled by proper choice of material, design, coatings, and occasionally by changing the environment. Various types of metallic and nonmetallic coatings are regularly used to protect metal parts from corrosion.

Identification of the metal or metals, environment the metal was subjected to, foreign matter and/or surface layer of the metal is beneficial in failure determination. Examples of some common types of corrosion:

- Uniform corrosion
- Pitting corrosion
- Intergranular corrosion
- Crevice corrosion
- Galvanic corrosion
- Stress corrosion cracking

Creep Rupture (High Temperature Failure Analysis)

High temperature progressive deformation of a material at constant stress is called creep. High temperature is a relative term that is dependent on the materials being evaluated. Creep occurs under load at high temperature. Failures involving creep are usually easy to identify due to the deformation that occurs. Failures may appear ductile or brittle. Cracking may be either trans granular or intergranular. While creep testing is done at constant temperature and constant load actual components may experience damage at various temperatures and loading conditions.

Ductile and Brittle Metal Characteristics

- Ductile metals experience observable plastic deformation prior to fracture.
- Brittle metals experience little or no plastic deformation prior to fracture. At times metals behave in a transitional manner - partially ductile/brittle.
- Ductile fracture has dimpled, cup and cone fracture appearance. The dimples can become elongated by a lateral shearing force, or if the crack is in the opening (tearing) mode.
- Brittle fracture displays either cleavage (trans granular) or intergranular fracture. This depends upon whether the grain boundaries are stronger or weaker than the grains.
- The fracture modes (dimples, cleavage, or intergranular fracture) may be seen on the fracture surface, and it is possible all three modes will be present of a given fracture face.

Hydrogen Embrittlement

- When tensile stresses are applied to a hydrogen embrittled component it may fail prematurely. Hydrogen embrittlement failures are frequently unexpected and sometimes catastrophic. An externally applied load is not required as the tensile stresses may be due to residual stresses in the material. The threshold stresses to cause cracking are commonly below the yield stress of the material.
- High strength steel, such as quenched and tempered steels or precipitation hardened steels are particularly susceptible to hydrogen embrittlement. Hydrogen can be introduced into the material in service or during materials processing.
- Hydrogen Embrittlement Failures
- Very small amounts of hydrogen can cause hydrogen embrittlement in high strength steels. Common causes of hydrogen embrittlement are pickling, electroplating, and welding; however, hydrogen embrittlement is not limited to these processes.
- Hydrogen embrittlement is an insidious type of failure as it can occur without an externally applied load or at loads significantly below yield stress. While high strength steels are the most common case of hydrogen embrittlement all materials are susceptible.

Examples of hydrogen induced damage are:

- formation of internal cracks, blisters, or voids in steels.
- embrittlement (i.e., loss of ductility).
- high temperature hydrogen attack (i.e., surface decarburizing and chemical reaction with hydrogen).

Prevention or Remedial Action

- internal cracking or blistering
    - use of steel with low levels of impurities (i.e., sulfur and phosphorus).
    - modifying environment to reduce hydrogen charging.
    - use of surface coatings and effective inhibitors.
- hydrogen embrittlement
    - use of lower strength (hardness) or high resistance alloys.
    - careful selection of materials of construction and plating systems.
    - heat treatment to remove absorbed hydrogen.
- high temperature hydrogen attack
    - selection of material (for steels, use of low and high alloy Cr-Mo steels; selected Cu alloys; non-ferrous alloys).
    - limit temperature and partial pressure H2.

Liquid Metal Embrittlement Failure

- Liquid metal embrittlement is the decrease in ductility of a metal caused by contact with liquid metal.
- The decrease in ductility can result in catastrophic brittle failure of a normally ductile material. Very small amounts of liquid metal are sufficient to result in embrittlement.
- The liquid metal can not only reduce the ductility but significantly reduce tensile strength. Liquid metal embrittlement is an insidious type of failure as it can occur at loads below yield stress. Thus, catastrophic failure can occur without significant deformation or obvious deterioration of the component.
- Intergranular or trans granular cleavage fracture are the common fracture modes associated with liquid metal embrittlement. However, reduction in mechanical properties due to decohesion can occur. This results in a ductile fracture mode occurring at reduced tensile strength. An appropriate analysis can determine the effect of liquid metal embrittlement on failure.
- Some events that may permit liquid metal embrittlement under the appropriate circumstances are listed below:
    - Brazing
    - Soldering

- Welding
- Heat treatment
- Hot working
- Elevated temperature service

Residual Stresses

- Residual stresses can be sufficient to cause a metal part to suddenly split into two or more pieces after it has been resting on a table or floor without external load being applied. While this is not a common occurrence, experienced people in the metal working industry have witnessed this phenomenon. While there may be additional factors causing this to occur residual stresses help explain these occurrences.
- Residual stresses are stresses that are inside or locked into a component or assembly of parts. The internal state of stress is caused by thermal and/or mechanical processing of the parts. Common examples of these are bending, rolling, or forging a part. Another example are the thermal stresses induced when welding.
- Residual stresses can play a significant role in explaining or preventing failure of a component at times. One example of residual stresses preventing failure is the shot peening of component to induce surface compressive stresses that improve the fatigue life of the component.
- Unfortunately, there are also processes or processing errors that can induce excessive tensile residual stresses in locations that might promote failure of a component.
- It must be kept in mind that the internal stresses are balanced in a component. Tensile residual stresses are counterbalanced by compressive residual stresses. To better visualize residual stresses, it is sometimes helpful to picture tension and compression springs to represent tensile and compressive residual stresses. While this is an aid to understanding, it must be kept in mind that residual stresses are three-dimensional.
- Residual stresses can result in visible distortion of a component. The distortion can be useful in estimating the magnitude or direction of the residual stresses.
- Thermal residual stresses are primarily due to differential expansion when a metal is heated or cooled. The two factors that control this are thermal treatment (heating or cooling) and restraint. Both the thermal treatment and restraint of the component must be present to generate residual stresses.
- A good common example of mechanically applied residual stresses is a bicycle wheel. A bicycle wheel is a very light and strong because of the way in which the components are stressed. The wire spokes are aligned radially and tightening the spokes creates tensile radial stresses. The spokes pull the rim inward, creating circumferential compression stresses in the rim. Conversely, the spokes pull the tubular hub outward. If the thin spokes were not under a proper tensile preload the thin wire spokes could not adequately support the load of the rider.

Failure Modes

- A failure mode is a characterization of the way a product failed.
- Failure Mode and Effects Analysis (FMEA): is a Bottom-Up Approach that examines potential failures in products or processes. It may be used to evaluate risk management priorities for mitigating known threat-vulnerabilities.
    - FMEA helps select remedial action that reduce cumulative impacts of life-cycle consequences (risks) from a systems failure (fault).
- Hazard Tree Analysis (HTA): is a Top-Down Approach starting with failures followed by diagnosis of the cause of failure to facilitate visual learning, this method illustrates connections between multiple contributing causes and cumulative (life-cycle) consequences.